

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	2.23%	7.21%	6.87%	1.89%
Meat MCal Target	20,360.80	65,798.92	62,676.71	17,238.59
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	320.60	399.69	83.51	271.44
MCals from meat	20360.80	65798.92	62676.71	17238.59
Max Utilized Mcals allowed from Dairy	21246.40	0.00	42389.85	17222.92
MCals actually Utilized from Dairy	21246.40	0.00	42389.85	17222.92
Percent of diet from their meat	2.23%	7.21%	6.87%	1.89%
Percent of Diet from their Dairy	2.33%	0.00%	4.65%	1.89%
Kg Fleece	2,424.00	0.00	0.00	0.00
1 Year+ (age) skins	119.43	15.49	16.22	101.16
Yearly Offspring skins	583.87	471.45	123.36	494.43
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	572.51	73.00	122.81	484.72
Males Step 1	160.30	199.84	41.76	135.72
Females Step 1	160.30	199.84	41.76	135.72
Replacement Mothers per year	114.80	14.90	15.54	97.24
Breeding Males Required	30.06	3.83	6.45	25.45
Additional Mothers	470.12	13.11	58.60	398.19
Males Step 2	131.63	35.88	19.92	111.49
Females 2	131.63	35.88	19.92	111.49
Total Mothers	687.31	87.90	138.35	581.96
Total Males	291.93	235.73	61.68	247.21
Total Females	28.67	163.96	21.83	24.23
Total Offspring	583.87	471.45	123.36	494.43
Total Animals	1,037.97	491.42	228.31	878.85
Units per Hectare	1.00			
Total Adults per Hectare	1.84	0.28	0.45	1.84
Total Offspring per Hectare	0.90	1.24	0.27	0.90
	1.54%		4.33%	1.30%
Total Adults	1,187	105	203	1,006
Total Offspring	584	471	123	494
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	103.44	0.00	14.99	87.53
Other	9,341.91	0.00	0.00	7,910.86
MCals Dry Mothers	120,815.78	0.00	31,737.50	102,235.65
total Maintenance MCals	130,157.69	0.00	31,737.50	110,146.50

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	347,694	215,242	206,498	294,432
Lactating	663,334	76,517	222,041	561,720
Maintenance	130,158	0	31,737	110,147
Adult Males	35,106	0	13,649	29,723
Offspring	160,272	473,812	451,484	135,721
TOTAL	1,336,564.5	765,571.1	925,410.3	1,131,742.4
Animal Diet Breakdown				
	Sheep	Pig	Cow	Goat
Crop	Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0
MCal kg ⁻¹				
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0
MCal kg ⁻¹				
Days	90.0	90.0	90.0	90.0
Number of Animals	583.9	471.5	123.4	494.4
MCals from Straw/Chaff	74618.5	0.0	78131.0	63188.0
MCals from Grain	118233.6	95469.5	49959.3	100121.8
MCals from other feed	192852.1	95469.5	128090.3	163309.7

Diet Calculations		
Plants Mcal	649,140.828	
Meat Mcal	182,500	
Dairy	80,859.172431353	
	Perfect	
Plants %	71.14%	
Meat %	20.00%	
Dairy %	8.86%	
check	Perfect	
Percent		
Domestic Meat	18.20%	
Dairy	8.86%	
Wild Meat	1.80%	
Crops	64.02%	
Wild Plants	7.11%	
check	Perfect	
new dairy	80859.17	
potential dairy < limit	TRUE	
new limit	80.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	166,075.000	
Dairy	80,859.172	
Wild Meat	16425.000	
Crops	584,226.745	
Wild Plants	64,914.083	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd	687	88	138	582				
Males for Slaughter	292	236	62	247				
Females for Slaughter	29	164	22	24				
Offspring available for Consumption	584	471	123	494				
Gestation Req'd	190919	182477	139796	161643				
Lactation Req'd	364238	64869	150319	308384				
Maintenance Req'd	433440	0	41500	367028				
Adult Males	35106	0	13649	29723				
Offspring Required	191801	516242	540300	162420				
Energy Required	1215505	763588	885564	1029198				
Ewe Fleece Kg Yr-1	1546							
Ram Fleece Kg year-1	60							
Juvenile Fleece Kg Yr-1	817							
Kg Fleece Year-1	2424	0	0	0				